Carbon footprint of Textile Heritage products

Carbon Footprint of Negamam Cotton Saree

1. Background

This study evaluates the carbon footprint (CF) of Negamam cotton sarees, a traditional handloom textile produced in Negamam village, located in the Pollachi region of Coimbatore district, Tamil Nadu, India. These sarees are known for their durability, dense weave structure, and cultural significance within the region. Historically woven using 80s combed cotton yarn and characterized by high fabric density (70–76 PPI), Negamam sarees represent a unique blend of craftsmanship, climatic adaptation, and cultural continuity.

The production of Negamam sarees is deeply rooted in the Devanga weaving community, with knowledge transmitted across generations through household-based handloom practices. Unlike mechanized textile production, the process relies heavily on manual labour and traditional techniques such as natural sizing, pit loom weaving, and vat dyeing.

ongoing initiatives aim to revive and sustain this heritage craft through improved process understanding and sustainability assessment. In this context, quantifying the carbon footprint becomes essential to identify environmental hotspots and support sustainable interventions in traditional textile systems.

2. Goal and Scope

The primary goal of this study is to quantify the greenhouse gas (GHG) emissions associated with the production of a Negamam cotton saree using a Life Cycle Assessment (LCA) approach.

The study adopts a **gate-to-gate system boundary**, covering all processes from yarn procurement to the finished saree. The analysis includes key production stages such as scouring, dyeing, washing, warping, sizing, weaving, finishing, and packaging. It also accounts for resource consumption (water, fuel, electricity), chemical usage, transportation, and waste generation within the production system.

The scope excludes upstream processes such as cotton cultivation, ginning, and spinning, as well as downstream stages including consumer use and end-of-life disposal.

3. Functional Unit

The functional unit (FU) for this study is defined as:

One finished Negamam cotton saree weighing approximately 530 grams, with dimensions of 1.17 m × 6.20 m.

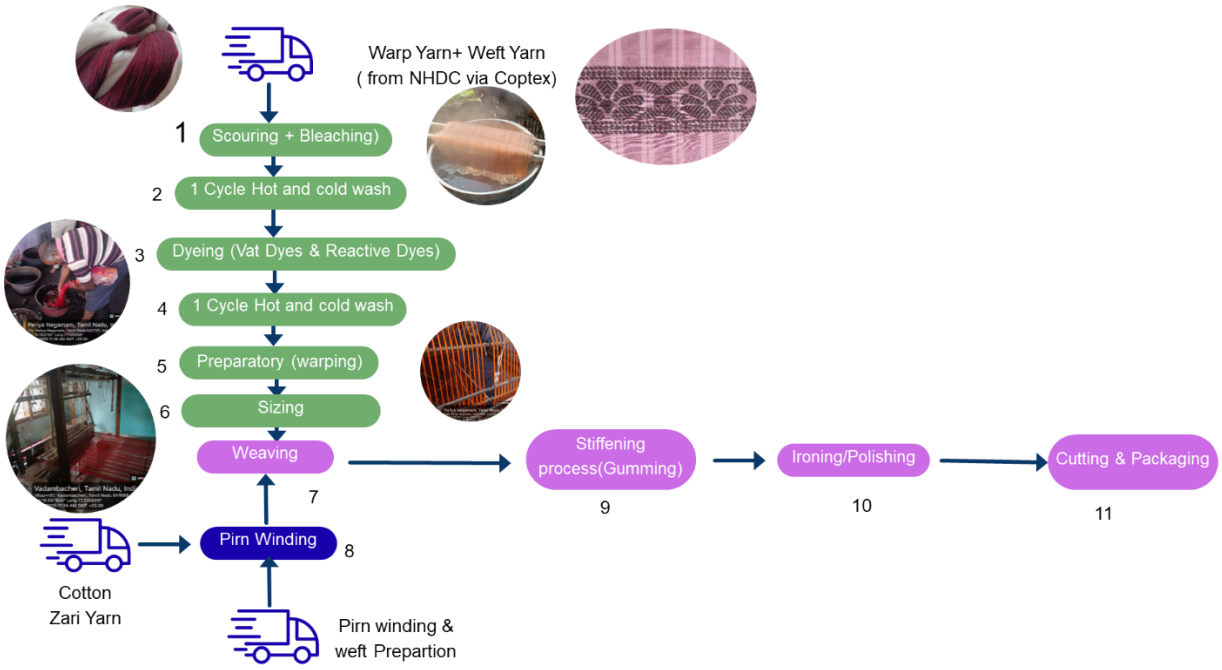

Figure Process tree for negamam saree.

4. Interpretation of Results

The results of the carbon footprint assessment indicate that the production of one Negamam cotton saree generates approximately 5.35–5.83 kg CO_2-equivalent per functional unit, reflecting the cumulative greenhouse gas emissions associated with the gate-to-gate production system. This value is characteristic of handloom-based textile systems, where emissions are primarily concentrated in pre-loom and wet processing stages rather than in the weaving process itself.

A detailed examination of process-level contributions reveals that the dyeing and sizing stages dominate the overall carbon footprint, accounting for nearly 77% of total emissions. This significant contribution can be attributed to the intensive use of thermal energy and chemicals during these processes. The use of firewood for heating during dyeing and sizing significantly

increases emissions. Although it is a traditional and easily available fuel, its inefficient burning leads to higher environmental impact. The weaving stage contributes a relatively small share of emissions (approximately 9–10%).

From an input perspective, the analysis highlights that fuel consumption—especially firewood—is the single largest contributor to greenhouse gas emissions, accounting for over 80% of the total impact.

Overall, the results demonstrate that the environmental performance of Negamam saree production is strongly influenced by energy use patterns in wet processing stages rather than the weaving process itself. This suggests that targeted interventions in energy sourcing, process efficiency, and chemical management could lead to substantial reductions in carbon emissions.

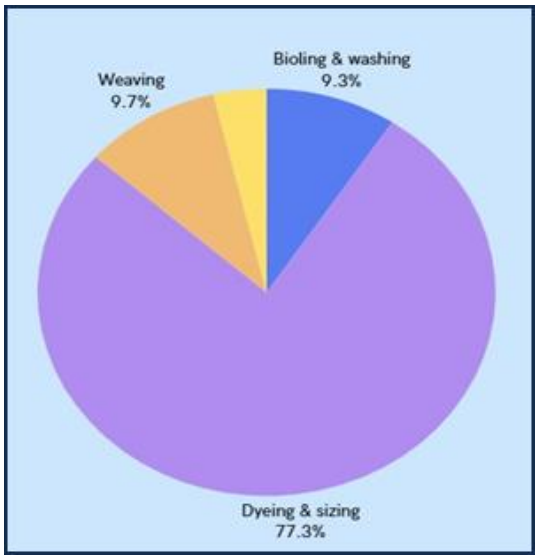

Figure Carbon footprint of 1 Negamam saree.

Carbon Footprint of Baluchari Silk Saree

1. Background

This study evaluates the carbon footprint (CF) of Baluchari silk sarees, a renowned heritage textile produced in Bishnupur, West Bengal, India. These sarees are distinguished by their intricate jacquard-woven motifs that depict mythological narratives, socio-cultural scenes, and historical events. Traditionally woven using mulberry silk yarn and zari, Baluchari sarees represent a unique confluence of artistic expression, cultural heritage, and technical craftsmanship.

The origins of Baluchari weaving trace back to the Murshidabad region during the early 18th century, with subsequent migration of artisans to Bishnupur following environmental and socio-political changes. Over time, the craft evolved from natural dyeing techniques to the use of synthetic dyes and from purely hand-operated looms to jacquard-based weaving systems.

The production system is characterized by high labour intensity, long weaving durations (typically 7–10 days per saree), and dependence on skilled artisans. However, challenges such as low wages, competition from power looms, and declining interest among younger generations threaten the sustainability of this craft. In this context, assessing the carbon footprint of Baluchari saree production is essential.

2. Goal and Scope

The primary goal of this study is to quantify the greenhouse gas (GHG) emissions associated with the production of a traditional Baluchari silk saree using a Life Cycle Assessment (LCA) approach.

A gate-to-gate system boundary is adopted, encompassing all processes from raw silk yarn input to the finished saree. The assessment includes degumming, dyeing, washing, drying, winding, warping, drafting, denting, weaving using jacquard looms, and finishing processes such as polishing and packaging. Resource inputs such as electricity, water, fuel (LPG), chemicals (acid dyes, acetic acid, soda ash), and transportation of raw materials are considered within the system boundary.

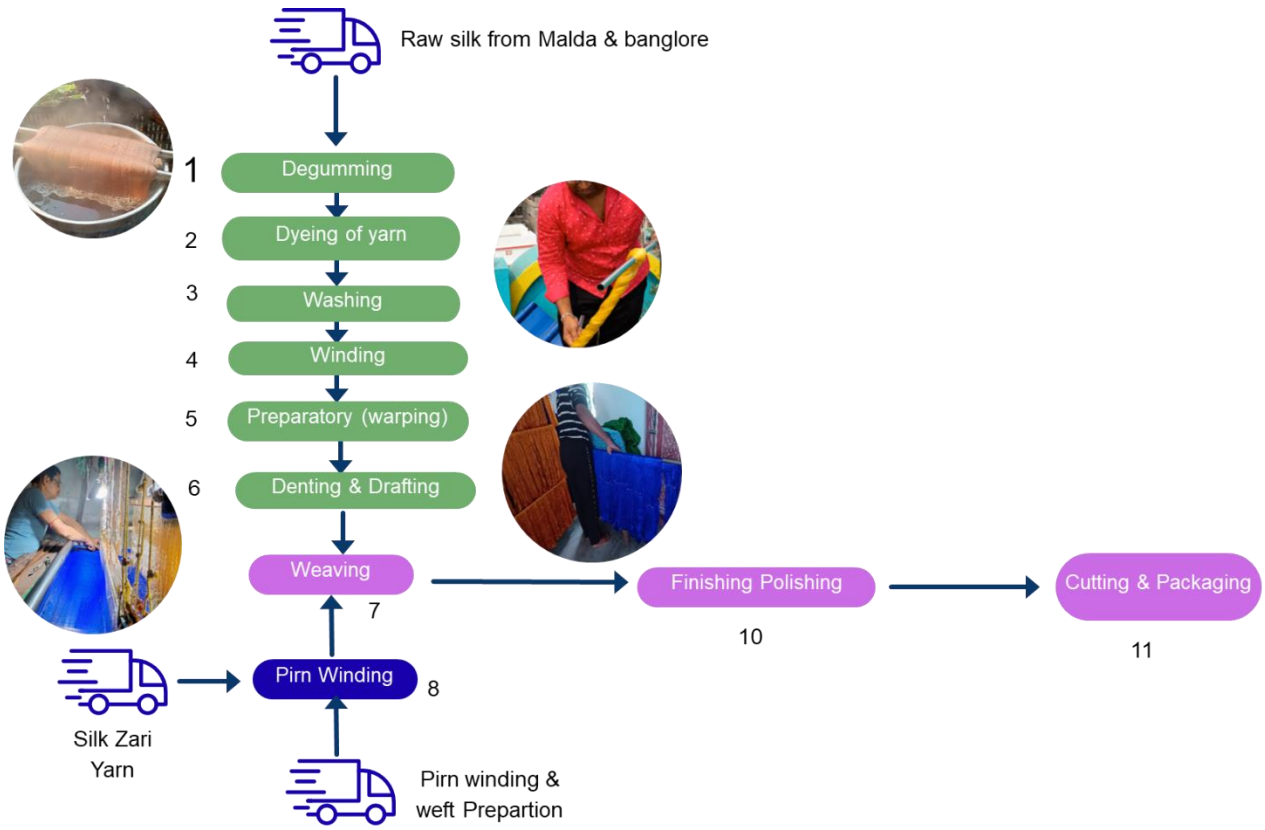

Figure Process tree for baluchari saree

3. Functional Unit

The functional unit (FU) for this study is defined as:

One finished Baluchari silk saree weighing approximately 650 grams, with dimensions of 1.1 m × 6.5 m.

4. Interpretation of Results

The carbon footprint assessment of the Baluchari silk saree indicates that the production of one saree results in approximately 3.99 kg CO_2-equivalent per functional unit, representing the cumulative greenhouse gas emissions within the defined gate-to-gate system boundary. This value reflects the environmental performance of a silk-based handloom product, where both energy use and material processing contribute to overall emissions.

During examination of the results shows that electricity consumption is the dominant contributor, accounting for nearly 72% of total emissions. This high contribution is primarily associated with energy-intensive processes such as yarn preparation, jacquard-based weaving, and auxiliary operations including winding and finishing.

The second major contributor is wastewater generation, contributing approximately 19% of total emissions. This is primarily linked to wet processing stages such as degumming, dyeing, and washing of silk yarns. These processes require substantial water input and involve

chemicals such as acid dyes, acetic acid, and soda ash, leading to wastewater streams with environmental implications.

Overall, the results indicate that the environmental performance of Baluchari saree production is primarily driven by energy use in electrically powered processes and wet processing stages, rather than by manual weaving activities alone..

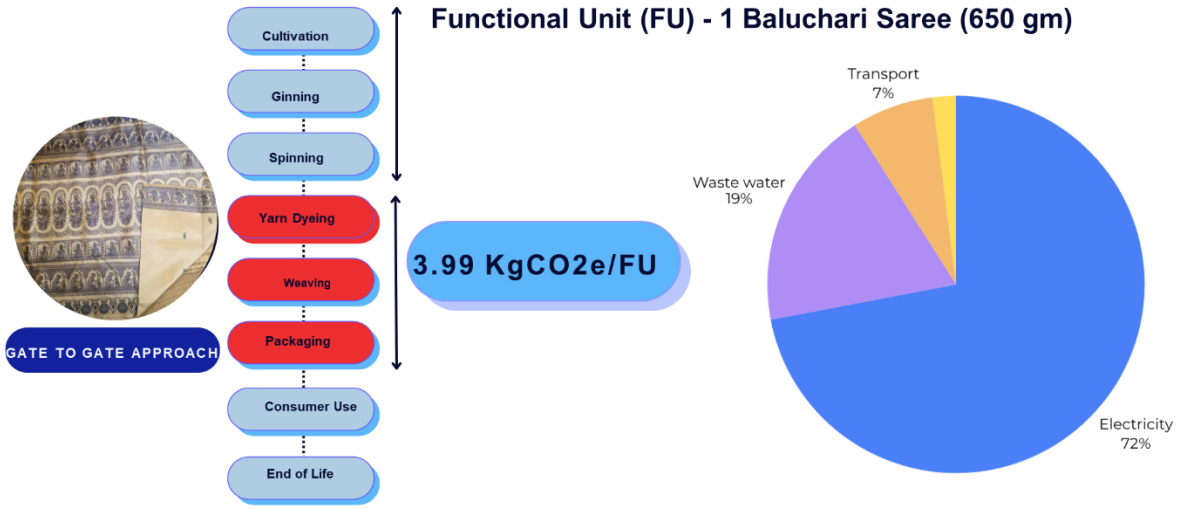

Figure Carbon footprint of 1 Baluchari saree.

Carbon Footprint of Muslin Cotton Fabric

1. Background

This study evaluates the carbon footprint (CF) of Muslin fabric, one of the finest and most historically significant textile traditions of the Indian subcontinent, particularly Bengal. Often described as the “woven wind” due to its exceptional fineness, transparency, and lightweight nature, Muslin represents a unique combination of craftsmanship, material science, and cultural heritage. Traditionally produced using ultra-fine cotton yarns (often exceeding 300 counts), Muslin fabric is characterized by its plain weave structure, breathability, and delicate texture.

The production of Muslin fabric remains largely manual, involving hand ginning, hand spinning using charkha, and handloom weaving. This highly skill-intensive process depends on fine motor control and favourable environmental conditions, such as humidity, to prevent yarn breakage. Despite its minimal mechanization, the production system faces challenges related to low productivity, labour-intensive operations, and economic viability.

In this context, assessing the carbon footprint of Muslin fabric provides valuable insights into its environmental performance and sustainability potential.

2. Goal and Scope

The primary goal of this study is to quantify the greenhouse gas (GHG) emissions associated with the production of Muslin cotton fabric using a Life Cycle Assessment (LCA) approach.

A gate-to-gate system boundary is adopted, covering all processes from raw cotton input to the finished Muslin fabric. The assessment includes stages such as hand ginning, carding, spinning, yarn formation, sizing, warping, weaving, and finishing.

2. Functional Unit

The functional unit (FU) for this study is defined as:

1 kg of finished Muslin white cotton fabric (approximately 35 GSM, yarn count 180s warp × 120s weft).

4. Interpretation of Results

The carbon footprint assessment of Muslin fabric indicates that the production of 1 kg of finished fabric results in approximately 3.84 kg CO_2-equivalent per functional unit within the gate-to-gate system, with significantly higher impacts observed when upstream processes are considered

A detailed analysis shows that electricity consumption is the dominant contributor, accounting for approximately 96% of total emissions within the system boundary. This contribution is mainly associated with auxiliary processes such as limited mechanized spinning (in some cases using solar/DC-powered systems), lighting, and support operations. Although Muslin production is largely manual, even small amounts of electricity use significantly influence the overall carbon footprint due to the otherwise low energy baseline of the system.

In contrast, water use and wastewater generation contribute only around 3% of total emissions, reflecting the minimal reliance on wet processing in traditional Muslin production. Processes such as bleaching and dyeing are either limited or performed using low-energy techniques, such as cold dyeing methods, which reduces the environmental burden compared to conventional textile processing systems.

The results also indicate that material losses, particularly cotton waste during ginning and spinning, are substantial, although their direct contribution to carbon emissions remains

relatively low within the defined system boundary. However, from a resource efficiency perspective, these losses highlight opportunities for improving material utilization.

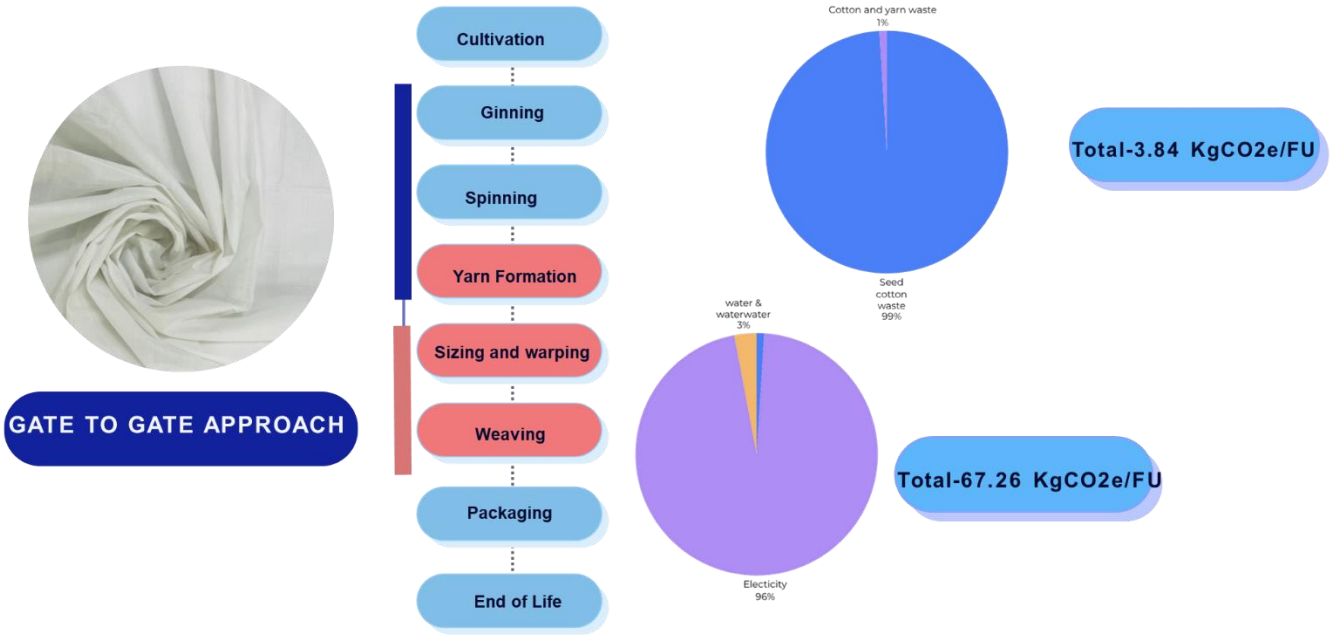

Figure Carbon footprint of 1 kg of finished Muslin white cotton fabric